Project 16

Australian Clinical Labs Online Reporting System

INTRODUCTION

The Diagnostic Lab Reporting System administrator program which will provide various analysis working on the internet. Here victims are first able to sign-up on the site and sign in using authorized information. Once authorized with their deal with and contact information, the patient may now see a wide range of assessments performed by the lab along with their expenses. The program allows for CBC, Blood Sugar, KFT, LFT assessments to be reserved by individual. The assessments also comprises of factors like Haemoglobin, WBC, etc. Now the program allows customers to book any analyze needed. After effective reservation program determines expenses and allows customers to pay online.

Objectives

Technology has facilitates human beings in almost every field of life. They turn manual tasks automatic to saves recourses. Automatic works is considered more trustful, reliable, accurate etc. Technology is the main reason, which successfully manages many processes and creates successful management system.

System Specifications

Existing Solution:

It needs employment as the human efforts are being automated by this system.

In this existing Diagnostic Lab Reporting System, the reports are generated either manually or by using computer software in which the details itself needs to be fitted by the employees of the organization. Existing system has greater tendency of having more errors as compared to the automation system. In existing Diagnostic Lab Reporting System, as the chances of errors are greater, thus the chances of getting information from the reports are more which will indirectly or directly affect the patients and their health only.

Existing Diagnostic Lab Reporting System needs to change asap in order to reduce the errors which affect the patient's report and indirectly patient's health in this way or any other way.

Proposed Solution:

The proposed system is an "Online Diagnostic Lab Reporting System". Its main aim is to bring together various diagnostic working, researches on one single platform that is also online (so that it is accessible for everyone).

- The system allows automate diagnosis system.
- Allows for faster service.
- Allows increased sales and profits for diagnostic labs.
- Easy, user-friendly GUI.

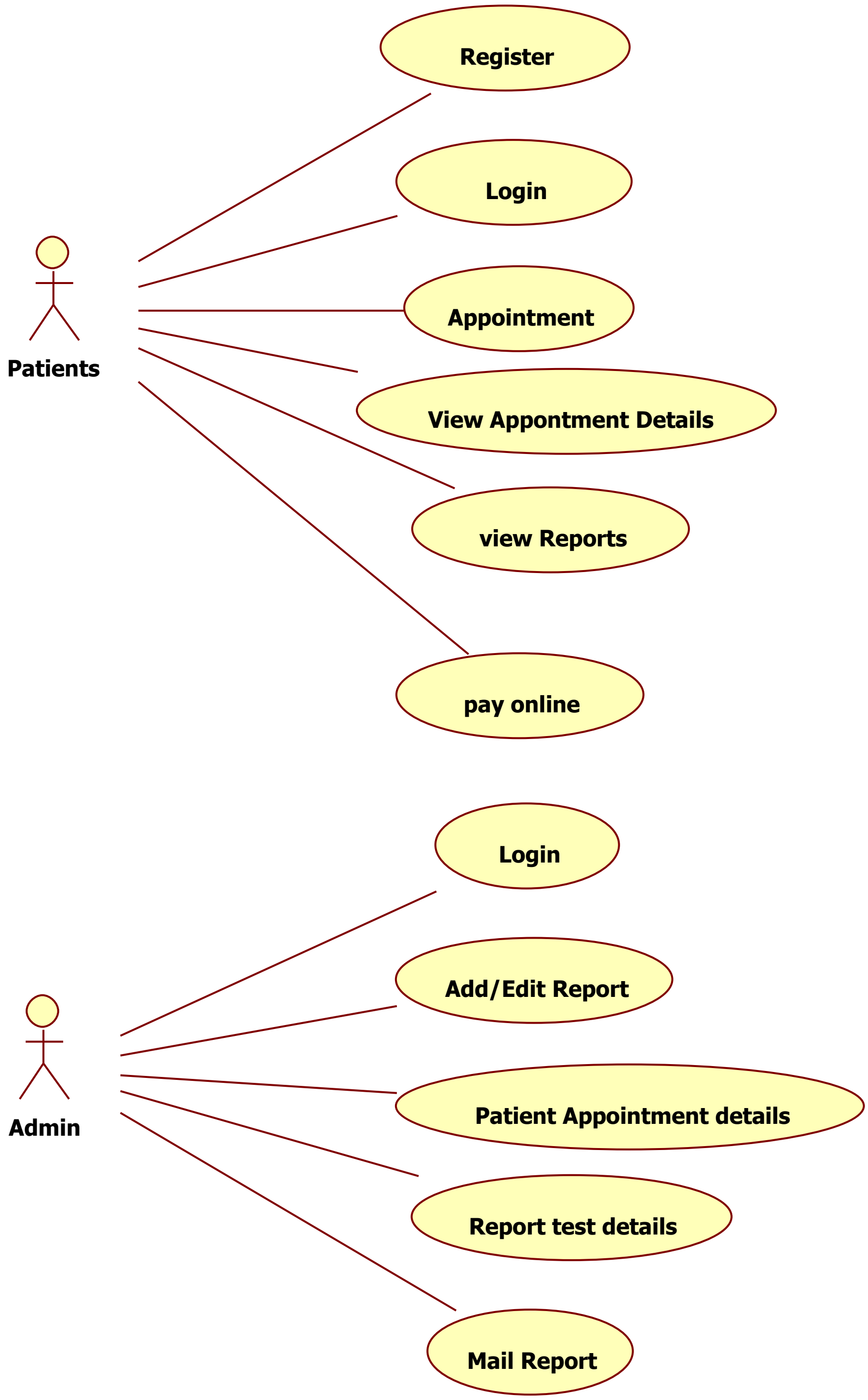
OVERALL DESCRIPTION OF THE PROPOSED SYSTEM

System Modules

→ User authentication and role-management module
→ Patient-management module
→ Clinician and referral module
→ Test catalogue module
→ Test-request module
→ Specimen collection and tracking module
→ Laboratory worklist module
→ Test-result management module
→ Result validation and authorisation module
→ Critical-result notification module
→ Clinical report-generation module
→ Clinician portal
→ Patient portal
→ Reporting and analytics module
→ Audit and system-administration module

Use case Diagrams:

Use case diagrams model behaviour within a system and helps the developers understand of what the user require. The stick man represents what's called an actor. Use case diagram can be useful for getting an overall view of the system and clarifying who can do and more importantly, what they can't do.



Use case diagram consists of use cases and actors and shows the interaction between the use case and actors.

- The purpose is to show the interactions between the use case and actor.
- To represent the system requirements from user's perspective.
- An actor could be the end-user of the system or an external system

Detailed Breakdown of Functional Requirements

Find below detailed breakdown of functional requirements:

FR1: User Registration

The system should allow authorised users, including clinicians, laboratory staff and patients, to register an account using verified personal and professional information.

FR2: User Authentication

The system should allow registered users to log in using a unique username or email address and password.

FR3: Multi-Factor Authentication

The system should support multi-factor authentication for laboratory staff, clinicians and system administrators.

FR4: Role-Based Access Control

The system should assign permissions according to user roles, including:

- Patient
- Referring clinician
- Reception staff
- Specimen collector
- Laboratory technician
- Laboratory scientist
- Pathologist
- Laboratory manager
- System administrator

FR5: User Account Management

Authorised administrators should be able to create, update, activate, suspend and deactivate user accounts.

FR6: User Profile Management

Users should be able to view and update permitted profile information, including contact details, organisation details and notification preferences.

FR7: Patient Registration

Authorised staff should be able to register a patient by recording:

- Full name
- Date of birth
- Sex or gender information where clinically relevant
- Contact details
- Address
- Patient identifier
- Medicare or insurance details where applicable
- Emergency contact information

FR8: Patient Identity Verification

The system should verify patient identity using multiple identifiers, such as full name, date of birth and patient identification number.

FR9: Patient Record Search

Authorised users should be able to search for patient records using name, patient identifier, date of birth, phone number or referral number.

FR10: Duplicate Patient Detection

The system should detect possible duplicate patient records and alert authorised staff before a new record is created.

FR11: Laboratory Test Catalogue

The system should maintain a catalogue of available laboratory tests, including:

- Test name
- Test code
- Specimen type
- Collection requirements

- Reference range
- Expected completion time
- Preparation instructions
- Test price where applicable

FR12: Test Request Creation

Referring clinicians and authorised staff should be able to create laboratory test requests for registered patients.

FR13: Electronic Referral Management

The system should allow clinicians to submit electronic referrals containing:

- Patient details
- Requested tests
- Clinical notes
- Provisional diagnosis
- Medication information
- Referring clinician details
- Priority level

FR14: Test Request Validation

The system should check that mandatory patient, clinician and test information has been provided before accepting a request.

FR15: Test Request Priority

The system should allow test requests to be classified as:

- Routine
- Urgent
- Critical
- Time-sensitive

FR16: Test Request Modification

Authorised users should be able to modify or cancel a test request before testing begins.

The system should record the reason for each modification or cancellation.

FR17: Specimen Registration

The system should allow staff to register each collected specimen against the relevant patient and test request.

FR18: Barcode Generation

The system should generate a unique barcode or specimen identification number for each specimen.

FR19: Specimen Label Printing

The system should allow staff to print specimen labels containing the specimen identifier, patient identifiers, specimen type and collection date and time.

FR20: Specimen Collection Recording

The system should record:

- Collection date and time
- Collection location
- Collector identity
- Specimen type
- Number of specimen containers
- Special collection conditions

FR21: Specimen Tracking

The system should track each specimen through stages such as:

- Requested
- Collected
- Received
- Accessioned
- In testing
- Awaiting review
- Completed
- Reported
- Stored
- Disposed

FR22: Specimen Receipt

Laboratory staff should be able to confirm specimen receipt and record the receiving staff member, date, time and specimen condition.

FR23: Specimen Acceptance and Rejection

Authorised laboratory staff should be able to accept or reject a specimen.

For rejected specimens, the system should record:

- Rejection reason
- Date and time
- Responsible staff member
- Required follow-up action

FR24: Specimen Recollection Request

The system should allow laboratory staff to request specimen recollection and notify the responsible clinician or collection centre.

FR25: Laboratory Worklist Generation

The system should generate laboratory worklists based on:

- Test type
- Laboratory department
- Priority
- Specimen status
- Assigned analyser
- Collection date
- Expected completion time

FR26: Test Assignment

Laboratory managers or authorised scientists should be able to assign tests to laboratory staff, departments or laboratory analysers.

FR27: Test Result Entry

Authorised laboratory staff should be able to enter:

- Numerical results
- Text-based results
- Positive or negative results
- Observations
- Units of measurement
- Method information
- Comments

FR28: Automated Result Import

The system should support the import of test results from compatible laboratory analysers or approved external systems.

FR29: Result Validation Rules

The system should validate entered results against:

- Permitted data formats
- Measurement units
- Reference ranges
- Patient age
- Patient sex where clinically relevant
- Test-specific validation rules

FR30: Abnormal Result Identification

The system should automatically identify and clearly mark results that fall outside the applicable reference range.

FR31: Critical Result Alerts

The system should generate a high-priority alert when a result meets a defined critical-value threshold.

FR32: Critical Result Acknowledgement

Authorised staff should be required to record:

- Person notified
- Notification method
- Date and time
- Staff member making the notification
- Notification outcome
- Acknowledgement status

FR33: Result Review

Laboratory scientists or pathologists should be able to review results, previous results, clinical notes and related test information before authorisation.

FR34: Result Correction

Authorised users should be able to correct a result before or after release.

The system should retain the original result and record the correction reason, date, time and responsible user.

FR35: Result Authorisation

Only authorised laboratory scientists or pathologists should be able to approve and release laboratory results.

FR36: Clinical Comment Management

Authorised pathologists and laboratory scientists should be able to add interpretive comments, recommendations and follow-up advice to a laboratory report.

FR37: Laboratory Report Generation

The system should generate a clinical laboratory report containing:

- Laboratory details
- Patient identifiers
- Referring clinician details
- Specimen information
- Requested tests
- Test results
- Reference ranges
- Abnormal-result indicators
- Clinical comments
- Authorising professional
- Report date and time
- Report status

FR38: Preliminary and Final Reports

The system should support report statuses including:

- Preliminary
- Partially completed
- Final
- Corrected
- Amended
- Cancelled

FR39: Report Distribution

The system should distribute authorised reports to approved recipients through secure methods such as:

- Clinician portal
- Patient portal
- Secure email notification

- Approved clinical information system
- Printable report

FR40: Clinician Portal

The clinician portal should allow authorised clinicians to:

- Submit test requests
- Track request status
- View patient results
- Download reports
- View historical results
- Acknowledge critical notifications
- Request clarification from the laboratory

FR41: Patient Portal

The patient portal should allow verified patients to:

- View released laboratory reports
- Download reports
- Review test history
- View plain-language test information
- Manage notification preferences
- Access instructions provided by the laboratory

Results requiring clinician consultation may be withheld from the patient portal until authorised for release.

FR42: Result History and Trend Display

The system should allow authorised users to compare current and previous results for the same test.

The system may display trends using tables or graphs.

FR43: Notification Management

The system should send notifications for events including:

- Test request received
- Specimen received
- Specimen rejected
- Recollection required
- Report available
- Critical result identified
- Corrected report issued

- Delayed test processing

FR44: Audit Trail

The system should record important activities, including:

- Login attempts
- Patient record access
- Test requests
- Specimen updates
- Result entry
- Result modification
- Report authorisation
- Report access
- Report download
- Administrative changes

Each audit record should include the user, action, date, time and affected record.

FR45: Laboratory Reporting and Analytics

Authorised managers should be able to generate reports showing:

- Number of tests requested
- Number of tests completed
- Turnaround times
- Rejected specimens
- Recollection rates
- Critical results
- Delayed tests
- Tests by department
- Tests by referring organisation
- Staff workload
- Amended reports

Reports should support filtering by date, test type, department, location and status.

FR46: System Configuration Management

Authorised administrators should be able to configure:

- Test types
- Test codes
- Reference ranges
- Measurement units
- Critical-value thresholds
- User roles

- Laboratory departments
- Report templates
- Notification rules
- Specimen rejection reasons
- Test turnaround targets

Changes to clinical configuration settings should be recorded in the audit trail.

FR47: Compliance:

- The system should comply with relevant regulations and standards, such as HIPAA, ISO 15189, and other relevant standards in Australia.

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it's important to make them clear for the stakeholders. Generally, functional requirements describe system behaviour under specific conditions.

The developers of this system must enhance the performance and efficiency of the system by adding 15 to 20 more functional requirements. Students need to do their own research to find how they can improve the system and which FRs need to add. The group must need a prior approval from the stakeholders/project supervisor before finalizing these Functional Requirements.

These enhanced FRs must be reflected separately in Final SRS Report after the approval.

Non Functional Requirements

There are a lot of software requirements specifications included in the nonfunctional requirements of the system, which contains various processes, namely Security, Performance, Maintainability, and Reliability.

Security:

- Patient Identification: The system needs the patient to recognize herself or himself using the phone.
- Logon ID: Any users who make use of the system need to hold a Logon ID and password.

- Modifications: Any modifications like insert, delete, update, etc. for the database can be synchronized quickly and executed only by the ward administrator.
- Front Desk Staff Rights: The staff at the front desk can view any data in the system, and add new patients record to the HMS but they don't have any rights to alter any data in it.
- Administrator rights: The administrator can view as well as alter any information in the system.
- Cybersecurity Implementation: Identify ethical risks in database design and implement the actions of mitigation.
- Cybersecurity Implementation: Provide evidence that you have implemented the data encryption and anonymization of data.
- Cybersecurity Implementation: Perform ‘Data Protection Impact assessment’ to help ensure compliance, facilitate a privacy by-design approach and identify better practice.
- Cybersecurity Implementation: Implement the secure methods for data encryption, data security and data breach to maintain the privacy of end users.

Performance:

- Response Time: The system provides acknowledgment in just one second once the 'patient's information is checked.
- Capacity: The system needs to support at least 1000 people at once.
- User-Interface: The user interface acknowledges within five seconds.

- Conformity: The system needs to ensure that the guidelines of the Microsoft accessibilities are followed.

Maintainability:

- Back-Up: The system offers efficiency for data backup.
- Errors: The system will track every mistake as well as keep a log of it.

Reliability:

- Availability: The system is available all the time.

Project should aim at Business process automation.

- In computer system the person has to fill the various forms & number of copies of the forms should be easily generated at a time.
- In computer system, it is not necessary to create the manifest but we can directly print it, which saves time.
- To assist the staff in capturing the effort spent on their respective working areas.
- To utilize resources in an efficient manner by increasing their productivity through automation.
- The system s h o u l d generate types of information that can be used for various purposes.
- It satisfy the user requirement
- Be easy to understand by the user and operator
- Be easy to operate
- Have a good user interface
- Be expandable
- Delivered on schedule within the budget.

Hardware Requirement: Should be recommended by the developers.
Software Requirement: Should be recommended by the developers.